

Research Upgrade Map: ADAS for Indian Roads

Each row: the part of your system, what to upgrade, and the paper that supports it. Click the blue **Open Paper** button to open the paper.

Current part	Upgrade (in plain words)	Research paper	Year	Link
1. SENSING AND PERCEPTION				
Camera + LiDAR fusion	Fuse camera and LiDAR at object level, not just by stacking features. Helps small objects such as pedestrians.	IS-Fusion: Instance-Scene Collaborative Fusion for Multimodal 3D Object Detection CVPR 2024	2024	Open Paper
3D object detection	Use a faster sparse LiDAR detector that sees farther, so braking decisions start earlier.	SAFDNet: A Simple and Effective Network for Fully Sparse 3D Object Detection CVPR 2024 (Oral), via arXiv	2024	Open Paper
BEV representation	Build one stable bird's-eye-view map from the cameras to share with LiDAR, prediction and planning.	BEVNeXt: Reviving Dense BEV Frameworks for 3D Object Detection CVPR 2024 (camera-only paper)	2024	Open Paper
3D occupancy forecasting	Predict which space will be occupied next, even for object types your detector does not know.	Cam4DOcc: Camera-Only 4D Occupancy Forecasting CVPR 2024	2024	Open Paper
Unknown obstacles	Flag objects that have no class label (debris, animals) using LiDAR + camera.	Spotting the Unexpected (STU): 3D LiDAR Dataset for Anomaly Segmentation CVPR 2025	2025	Open Paper
Potholes	Estimate pothole size and depth by fusing camera detection with LiDAR depth.	Deep Learning Enhanced Feature Extraction of Potholes Using Vision and LiDAR Data IEEE Access 2024 (road-maintenance focus)	2024	Open Paper
2. TRACKING AND PREDICTION				

Current part	Upgrade (in plain words)	Research paper	Year	Link
Object tracking	Get cleaner speed and acceleration for every tracked object.	MCTrack: Unified 3D Multi-Object Tracking Framework arXiv preprint	2024	Open Paper
Multi-agent trajectory prediction	Predict several possible futures for each road user.	DeMo: Decoupling Motion Forecasting into Directional Intentions and Dynamic States NeurIPS 2024	2024	Open Paper
Interaction-aware prediction (pedestrians)	Retrain and test pedestrian prediction on unstructured traffic, where standard models perform worse.	INDIAN DATA Pedestrian Intention and Trajectory Prediction in Unstructured Traffic Using IDD-PeD arXiv preprint	2025	Open Paper
Interaction-aware prediction for planning	Keep only the vehicle interactions that matter to the planner. Faster MPC.	Scalable Multi-modal MPC via Duality-based Interaction Predictions arXiv preprint	2024	Open Paper
Uncertainty-aware prediction	Add calibrated safety zones around every predicted path.	Safe POMDP Online Planning among Dynamic Agents via Adaptive Conformal Prediction IEEE RA-L 2024	2024	Open Paper
3. RISK AND TTC				
TTC / collision prediction	Replace the fixed TTC formula with a learned, context-aware risk score.	Learning collision risk proactively from naturalistic driving data at scale (GSSM) arXiv preprint	2025	Open Paper
Predictive risk estimation	Calibrated risk over space and time to decide when to brake, with fewer false alarms.	Uncertainty-Aware Vision-based Risk Object Identification via Conformal Risk Tube Prediction arXiv preprint	2026	Open Paper
4. PLANNING AND CONTROL				

Current part	Upgrade (in plain words)	Research paper	Year	Link
Risk-aware planning	Plan several options (pass left, pass right, yield) under collision-probability limits.	A Vehicle System for Navigating Among Vulnerable Road Users arXiv preprint	2025	Open Paper
Dynamic local trajectory planning	Generate several candidate paths and keep a rule-based safety filter.	PLUTO: Pushing the Limit of Imitation Learning-based Planning arXiv preprint	2024	Open Paper
MPC / trajectory optimization	Plan that branches by predicted behavior, so it is less over-cautious.	Motion Planning under Uncertainty: Multi-Modal Predictors into Branch MPC arXiv preprint	2024	Open Paper
Adaptive speed, steering, braking	Choose the minimum-risk maneuver: brake, steer, or both.	Enhancing Pedestrian Safety Through Automated Emergency Steering IEEE T-ITS 2026	2026	Open Paper
5. EVALUATION AND INDIAN-ROAD DATA				
Closed-loop testing	Test the whole system in closed loop in CARLA, not just with open-loop error.	Bench2Drive: Closed-Loop End-to-End Autonomous Driving Benchmark NeurIPS 2024 (Datasets and Benchmarks)	2024	Open Paper
Indian multimodal data	Validate on Hyderabad data: 4 cameras, 6 radars, 1 LiDAR, GPS and IMU.	INDIAN DATA TIAND: A Multimodal Dataset for Autonomy on Indian Roads IEEE Xplore, June 2024	2024	Open Paper
Indian LiDAR data	Benchmark 3D detection and tracking on about 12k annotated LiDAR frames.	INDIAN DATA IDD-3D: Indian Driving Dataset for 3D Unstructured Road Scenes WACV 2023 (older, included as requested)	2023	Open Paper

Recommended Research Upgrade Path

Step 1	Camera + LiDAR fusion
Step 2	BEV and occupancy map (plus unknown obstacles and potholes)
Step 3	Tracking and multi-agent prediction
Step 4	Uncertainty estimation
Step 5	Predictive risk and collision probability
Step 6	Risk-aware local planning
Step 7	MPC and adaptive speed / steering / braking
Step 8	Closed-loop evaluation on Indian-style scenarios

Best 5 contributions for a student project

1. Calibrated uncertainty on Indian data (IDD-PeD, TIAND).
2. Learned risk instead of fixed TTC, tuned for two-wheelers.
3. Link prediction uncertainty to the MPC planner.
4. Unknown-obstacle and pothole layer as planner costs.
5. Closed-loop test set with Indian-style scenarios.

Notes: only the three papers marked INDIAN DATA use Indian road data; the rest were tested on Western or simulated data. The five contributions are my judgment, not claims from the papers. Papers labeled arXiv preprint are not confirmed as published elsewhere.

All Paper Links (backup list, click the title)

1. [IS-Fusion: Instance-Scene Collaborative Fusion for Multimodal 3D Object Detection](#)
2. [SAFDNet: A Simple and Effective Network for Fully Sparse 3D Object Detection](#)
3. [BEVNeXt: Reviving Dense BEV Frameworks for 3D Object Detection](#)
4. [Cam4DOcc: Camera-Only 4D Occupancy Forecasting](#)
5. [Spotting the Unexpected \(STU\): 3D LiDAR Dataset for Anomaly Segmentation](#)
6. [Deep Learning Enhanced Feature Extraction of Potholes Using Vision and LiDAR Data](#)
7. [MCTrack: Unified 3D Multi-Object Tracking Framework](#)
8. [DeMo: Decoupling Motion Forecasting into Directional Intentions and Dynamic States](#)
9. [Pedestrian Intention and Trajectory Prediction in Unstructured Traffic Using IDD-PeD](#)
10. [Scalable Multi-modal MPC via Duality-based Interaction Predictions](#)
11. [Safe POMDP Online Planning among Dynamic Agents via Adaptive Conformal Prediction](#)
12. [Learning collision risk proactively from naturalistic driving data at scale \(GSSM\)](#)
13. [Uncertainty-Aware Vision-based Risk Object Identification via Conformal Risk Tube Prediction](#)
14. [A Vehicle System for Navigating Among Vulnerable Road Users](#)
15. [PLUTO: Pushing the Limit of Imitation Learning-based Planning](#)
16. [Motion Planning under Uncertainty: Multi-Modal Predictors into Branch MPC](#)
17. [Enhancing Pedestrian Safety Through Automated Emergency Steering](#)
18. [Bench2Drive: Closed-Loop End-to-End Autonomous Driving Benchmark](#)
19. [TIAND: A Multimodal Dataset for Autonomy on Indian Roads](#)
20. [IDD-3D: Indian Driving Dataset for 3D Unstructured Road Scenes](#)